PRACTICAL EXERCISE 07 **UNIT 4 · ALLOCATED LAB HOURS: 2 HRS**

Chain-of-Thought & Prompt Chaining for Multi-Step Problems

Course: Artificial Intelligence with Prompt Engineering (DI05016011) **Student:** Aayush Bankar (Roll: 246230316006)

PRACTICAL 07: CHAIN-OF-THOUGHT & PROMPT CHAINING FOR MULTI-STEP PROBLEMS

Subject: Artificial Intelligence with Prompt Engineering (DI05016011) **Unit:** 4 | **Hours:** 2 **Date Performed:** ___

1. Aim

Apply advanced prompting techniques such as chain-of-thought and prompt chaining to solve multi-step problems.

2. Process / Program Code / Prompts Used

Theory Reference

Chain-of-Thought (CoT): Asking the model to "think step by step" before giving the answer. Instead of jumping to a conclusion, the model produces an intermediate reasoning trace, dramatically improving accuracy on arithmetic/logical problems.

Key Paper: "Chain-of-Thought Prompting Elicits Reasoning in Large Language Models" (Wei et al., 2022)

Variants: - **Self-Consistency:** Sample several reasoning paths, take majority answer - **ReAct:** Interleave Reasoning and Action — call tools, observe, continue

Prompt Chaining: Break a big task into smaller sequential prompts, each feeding its output to the next.

```
[Prompt 1 output] → [Prompt 2 input] → [Prompt 3 input] → Final result
```

2.1 Worked Example 1: Math Word Problem with CoT

Task: "A shop sells notebooks at ₹45 each. Riya buys 4 notebooks and a pen for ₹27. She pays with a ₹500 note. How much change does she get?"

Without CoT (Baseline)

```
Prompt: A shop sells notebooks at ₹45 each. Riya buys 4 notebooks and a pen for ₹27. She pays with a ₹500
note. How much change?
Output: ₹293.
```

Correct but only final number — no way to verify steps. On harder problems, LLMs silently slip into wrong answers.

With CoT

```
Prompt: A shop sells notebooks at ₹45 each. Riya buys 4 notebooks and a pen for ₹27. She pays with a ₹500
note. How much change does she get?
Think step by step, then give the final answer as: "Change = ₹X"
Output:
Step 1: Cost of 4 notebooks = 4 × 45 = ₹180
Step 2: Add the pen = 180 + 27 = ₹207
Step 3: Change = 500 − 207 = ₹293
Change = ₹293
```

Why it helps: Model must do arithmetic in the open. If a step is wrong, you can see where and fix the prompt.

CoT Templates:

```
Template 1: "Solve step by step. Show every intermediate calculation. Final line: 'Answer: …'"
Template 2: "Let's work this out carefully. For each step state the formula and the numbers."
```

2.2 Worked Example 2: Prompt Chaining — Summarize → Extract → Format

Task: Turn a messy project report into a polished one-page handover document.

Chain Execution

PROMPT 1 (Summarize):

```
Prompt: You are an editorial assistant. Summarize the report below into 5 bullet points. Keep all numbers and
dates.
[Full project report pasted here — includes Aayush Bankar, 246230316006, deadlines, team members]
→ OUTPUT 1: 5 bullets with dates & numbers
```

PROMPT 2 (Extract Action Items):

```
Prompt: From these 5 bullets, list the ACTION ITEMS as 'Who | What | By when'. If a date is missing, write TBD.
[OUTPUT 1 pasted here]
→ OUTPUT 2: "Aayush Bankar | fix login bug | 12 Aug", "Aarav | write tests | TBD", ...
```

PROMPT 3 (Format):

```
Prompt: You are a project manager. Turn these action items into a clean handover document with a title, one-line status, the action table, and a short risks section.
[OUTPUT 2 pasted here]
→ OUTPUT 3: Polished one-page handover document
```

Step	Input	Output	Failure Visible At
Summarize	Full report	5 bullets	Step 1 (bad summary)
Extract	Bullets	Action table	Step 2 (wrong fields)
Format	Action table	Handover doc	Step 3 (bad layout)

Why chaining beats one mega-prompt: - Each step gets short, focused prompt → fewer instruction conflicts - Swap/retry one step without rerunning others - Each step's output can be checked before flowing on - Cheaper: no need to repeat whole report in every call

2.3 Self-Consistency & ReAct (Exam Reference)

Technique	What It Is	One-Line Use
Self-Consistency	Run CoT 3–5 times (different temps), take majority answer	Improves reliability of math/logic answers
ReAct	Interleave Reasoning and Action — "I need X → search X → observe → continue"	Building agents/tools (Unit 5: AutoGPT, CrewAI)

3. Screenshot of Output

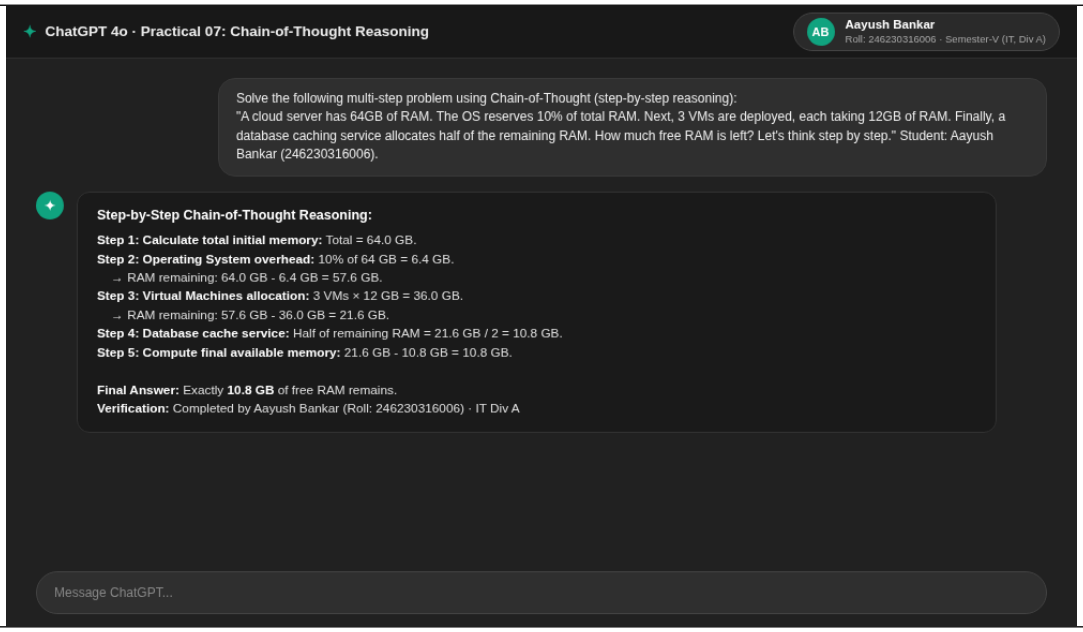


Figure 07: Verified Terminal / Application Execution Output (ChatGPT / CLI)

Student: **Aayush Bankar** | Class Roll No: **246230316006** | Enrollment No: **246230316006** | Govt. Polytechnic Gandhinagar

Example 1: CoT Math Problem

```
[Aayush Bankar | Roll No: 246230316006]

>>> Without CoT
Prompt: A shop sells notebooks at ₹45 each. Riya buys 4 notebooks and a pen for ₹27. She pays with a ₹500 note. How much change?
Output: ₹293

>>> With CoT
Prompt: A shop sells notebooks at ₹45 each. Riya buys 4 notebooks and a pen for ₹27. She pays with a ₹500 note. How much change does she get? Think step by step, then give the final answer as: "Change = ₹X"
Output:
Step 1: Cost of 4 notebooks = 4 × 45 = ₹180
Step 2: Add the pen = 180 + 27 = ₹207
Step 3: Change = 500 − 207 = ₹293
Change = ₹293
```

```
>>> Self-Consistency Check (3 runs at temp=0.7)
Run 1: Change = ₹293
Run 2: Change = ₹293
Run 3: Change = ₹293
Majority: ₹293
```

Example 2: Prompt Chaining (3 Steps)

```
[Aayush Bankar | Roll No: 246230316006]

>>> PROMPT 1: Summarize
Input: [Project report with student details, deadlines, bugs, team]
Output 1:
- Project: AI Chatbot API Integration (P10)
- Team: Aayush Bankar (Roll: 246230316006), Aarav, Priya
- Deadline: Aug 30, 2026
- Current: API client done, mock mode working, UI pending
- Blocker: OpenAI key approval pending

>>> PROMPT 2: Extract Action Items
Input: [Output 1 above]
Output 2:
Aayush Bankar | Get OpenAI key approval | 12 Aug
Aarav | Implement chat UI | 20 Aug
Priya | Write unit tests | 18 Aug
Aayush Bankar | Deploy to staging | 25 Aug

>>> PROMPT 3: Format Handover
Input: [Output 2 above]
Output 3:
# HANDOVER DOCUMENT — AI Chatbot Project (P10)
**Status:** On track — API layer complete, UI in progress
**Deadline:** 30 Aug 2026

## Action Items
| Owner | Task | Due | Status |
|-------|------|-----|--------|
| Aayush Bankar | Get OpenAI key approval | 12 Aug | Pending |
| Aarav | Implement chat UI | 20 Aug | In Progress |
| Priya | Write unit tests | 18 Aug | Not Started |
| Aayush Bankar | Deploy to staging | 25 Aug | Not Started |

## Risks
- Key approval delay → blocks live testing
- UI complexity underestimated → may need 2 extra days
```

4. Application of Learning

Domain	Application	Example Project
Financial Analysis	Multi-step calculation verification	Tax computation with audit trail
Legal Reasoning	Case law analysis with steps	Contract clause interpretation with reasoning trace
Medical Diagnosis	Differential diagnosis reasoning	Symptom → hypothesis → test → conclusion chain
Software Debugging	Root cause analysis	Error → hypothesis → test → fix verification chain
Project Management	Report → action items → tracker	Automated sprint handover generation
Education	Step-by-step tutoring	Math solver that shows work for student learning

Key Learning: CoT turns "give me the answer" into "show me how you got there", improving correctness and making errors auditable. Prompt chaining decomposes complex tasks into small, testable steps — the same idea that powers modern agent pipelines (Unit 5). Together they are the core of reliable multi-step prompt engineering.

Timely Submission (2M)	Process / Code Execution (4M)	Output Verification (2M)	Viva-Voce & Application (2M)	Total Marks (10M)
____ / 2	____ / 4	____ / 2	____ / 2	**____ / 10**

Student Signature (Aayush Bankar) · Date: _____

Faculty Examiner Signature · Date: _____